

Catching the Falling Ladoos!

An exciting year-end game project designed just for Grade 2.



Help Chotu the Monkey catch sweet treats from the sky!

Hi Young Coder! Today you will make your very own arcade game. You will learn how computers think, build the rules, and then get to play it yourself!

What We Will Learn Today!

Welcome to our coding adventure! Before we start building, let us look at the amazing things you are going to learn how to do today:

Our Fun Learning Goals:

- ★ **Score Boxes (Variables):** Learn how to make a box that keeps score when you catch a sweet!
- ★ **If-Then Rules (Logic):** Tell the computer: *"IF something happens, THEN do a fun action!"*
- ★ **Moving Pieces:** Bring together sounds, moving shapes, and screen buttons.
- ★ **Play and Share:** Have a blast playing your game and sharing it with your family!

By the end of this lesson, you will be the boss of your own computer screen. You will have a real game that you built with your own hands!

Concept 1: What is a Variable?

Imagine you have a little **cardboard box** on your desk. Inside this box, you can keep a number. Every time something happens, you can erase the old number and write a new one!

Think of Cricket!

Have you watched an IPL cricket match? On the screen, the **"Runs"** box keeps changing! Every time the batsman hits the ball and gets a run, the number in the box goes up. That box is a variable!

Our Game Score Box

In our game, we will make a variable box and name it **"Score"**. At first, the box will hold the number **0**. Every time you catch a delicious laddoo, we will tell the computer to add **1** to the box!

[Score Box] Currently holds: 0 → Catch a Ladoo! → Now holds: 1

Concept 2: If-Then Rules!

Computers need clear rules to know what to do. We call these rules **If-Then sentences**. You use these rules every single day without knowing it!



A Rule from Real Life:

***"IF** you finish your school homework by 6 PM,
THEN you can go outside to play badminton!"*

How Chotu Uses the Rule

In our arcade game, the computer is constantly checking for a specific secret moment. The rule we will code looks like this:

IF Chotu the Monkey touches a falling ladoo,
THEN make a funny POP sound and add points!

If Chotu misses the ladoo, nothing happens. But the exact moment they touch, the rule triggers the fun reaction!

Concept 3: Falling on the Grid

How does something move on a computer screen? The screen is like a giant, invisible map grid made of two magic paths: **X** and **Y**.

- **The X Path** goes from side to side (left to right).
- **The Y Path** goes up and down (from the sky to the grass).

The Falling Mango Trick

Think of a juicy Alphonso mango hanging on a high tree branch. When it gets ripe and breaks off, it drops straight down to the ground. As it falls closer to the dirt, its **Y position gets smaller and smaller!**

Making Our Ladoos Fall

To make our ladoos drop down from the top of the sky, we will tell the computer to keep changing the Y path by a minus number. This makes it sink downwards exactly like a ripe mango falling from a tree!

Step 1: Setting Up Your Game Canvas

Let's open Scratch on your computer and set up the stage for our game! Follow these simple steps to prepare your canvas:



1. Pick a Happy Background

Click on the backdrop button in the corner. Choose a bright, festive scene where you want your game to happen! You can pick a cozy **kitchen**, a magical **sweet shop**, or a colorful **village fair**.



2. Choose Your Game Characters (Sprites)

We need two characters on our stage. In coding, we call them sprites:

- **Catcher Catcher:** Add **Chotu the Monkey**. He will stand at the bottom of the screen.
- **The Prize:** Add a beautiful, round, golden **ladoo** (or a yellow circle ball). This will fall from the sky.

Step 2: Controlling Chotu

Now, we need to give you the power to steer Chotu the Monkey left and right using your keyboard arrow buttons! We do this by putting **If-Then** blocks inside a **Forever** loop.

Your Code Instructions:

Click on Chotu the Monkey sprite and build this script puzzle in the code zone:

When Green Flag Clicked

Forever Loop:

- ◆ **IF** Right Arrow Key is pressed? **THEN:**
Change X by 15 (Moves Chotu right!)
- ◆ **IF** Left Arrow Key is pressed? **THEN:**
Change X by -15 (Moves Chotu left!)

★ **Little Checkpoint:** Click the green flag above your stage! Tap your left and right arrow keys. Does Chotu glide smoothly across the floor? Great job!

Step 3: Falling Sweet Logic

Next, click on your **Ladoo sprite**. We need to make it drop down automatically, and reset at the top if it hits the bottom floor!

When Green Flag Clicked

Forever Loop:

👉 Change Y by -8 (This makes the sweet fall down continuously!)

👉 **IF** Y position is less than -170 (It touched the floor!) **THEN:**

Go to a Random X location

Set Y to 180 (Teleports it back up to the top sky!)

Why do we do this?

This clever trick keeps the game running forever! The ladoo falls down, drops off the bottom edge, disappears, and instantly flies back up to a brand-new random spot in the sky to surprise the player.

Step 4: Catching and Scoring!

This is the magic step where our game finally comes to life! We will connect Chotu and the sweet so you can score real points.



Making the Score Counter:

Go to the orange **Variables** circles in Scratch. Click the button that says *"Make a Variable"*. Type the name **Score** and click OK. Now you have a real live counter box on your screen!

The Catching Rule Code (Put this inside the Ladoo sprite):

IF Touching "Chotu the Monkey"? **THEN:**

✚ Change Score by 1 (Gives you a point!)

🎵 Play Sound "Pop" (Makes a happy audio crunch!)

🚀 Go to a Random X location and Set Y to 180 (Resets sweet back to top!)

💡 Remember: Make sure to add a block at the very start that resets your Score to 0 whenever the green flag is clicked!

Make the Game Your Own!

Hooray! You built a complete arcade game loop. Now you are a true game designer, and you can change the game elements to make it extra cool!

Fun Ideas for Customization:

- ★ **Swap the Food:** Don't want a ladoo? Draw a gooey **jalebi**, a piece of chocolate cake, or a cricket ball!
- ★ **Speed Burst:** Change the falling speed from -8 to -12 to make the sweets zoom down much faster!
- ★ **Win Condition:** Try adding a rule: If Score reaches 10, stop the game and celebrate!

 **Project Complete!** 

**Now call your mom, dad, or best friend over.
Show them your code blocks, hit full-screen, and let them play!**